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Computer Networks

CN101 | Suggested sequence · Semester 3 | Student-authored study guide

How to use these notes

Layered communication, addressing, routing, transport, DNS, HTTP, TLS, and network troubleshooting. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Layers, addressing, and routing

Layering separates responsibilities. IP addresses identify network interfaces; routers forward packets between networks. DNS maps names to records and caching affects how quickly changes propagate.

Worked example

```
A browser resolves a hostname, establishes a route to an IP address, then exchanges transport
and application data.
```

Check your understanding

- What does a router do?
- Why can DNS changes appear delayed?

MODULE 02 · Transport and application protocols

TCP provides an ordered byte stream with reliability and congestion control; UDP provides datagrams without those guarantees. HTTP defines request/response semantics; applications choose protocol behavior according to need.

Worked example

```
A web request identifies method, path, headers, and optionally a body; status code
communicates outcome.
```

Check your understanding

- What does TCP guarantee?
- What is the role of an HTTP status code?

MODULE 03 · TLS, security, and diagnosis

TLS protects data in transit and authenticates a server when certificate checks succeed. Use layered diagnostics: link, IP route, DNS, transport port, TLS handshake, and application response. Firewalls restrict permitted flows.

Worked example

```
If DNS resolves but a TCP connection times out, inspect routing/firewall/port before debugging
application JSON.
```

Check your understanding

- What does a TLS certificate help authenticate?
- What is a useful troubleshooting order?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.